



Carroll Dynamics in the Extended Carroll String

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Galilean String

The Gomis-Ooguri string sigma model was introduced in [1, 2] as a non-relativistic limit of the Polyakov action coupled to a constant Kalb-Ramond field in the longitudinal direction. Before taking the limit, we can write the action in the following form:

$$S = -\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} \left(\gamma^{ab} \partial_a X^{A'} \partial_b X_{A'} - c^2 \Gamma_+^{ab} \partial_a X^- \partial_b X^+ \right).$$

Where we have introduced longitudinal indices $A, B = 0, 1$ and transverse indices $A', B' = 2, \dots, N-1$ as well as the following notation:

$$\Gamma_{\pm}^{ab} = \gamma^{ab} \pm \varepsilon^{ab}, \quad X^{\pm} = X^0 \pm X^1.$$

In order to take the limit $c \rightarrow \infty$, we introduce auxiliary worldsheet fields λ_a and $\bar{\lambda}_a$ such that the divergent part of the action can be written in the following classically equivalent way:

$$-\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} \Gamma_+^{ab} \left(\lambda_a \partial_b X^+ + \partial_a X^- \bar{\lambda}_b + \frac{1}{c^2} \lambda_a \bar{\lambda}_b \right).$$

Therefore, we are able to take the limit $c \rightarrow \infty$, which gives the following non-relativistic action:

$$S_{\text{NR}} = -\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} \left(\gamma^{ab} \partial_a X^{A'} \partial_b X_{A'} + \Gamma_+^{ab} \lambda_a \partial_b X^+ + \Gamma_-^{ab} \bar{\lambda}_a \partial_b X^- \right).$$

Which is equivalent to the Gomis-Passerini model introduced in [3]. Upon going to the light-cone gauge, the astute reader will recognize this as the well-known Gomis-Ooguri model [1, 2]:

$$S_{\text{GO}} = T \int d^2\sigma \left(\partial_+ X^{A'} \partial_- X_{A'} + \lambda^- \partial_- X^+ + \lambda^+ \partial_+ X^- \right).$$

The equations of motion for the Gomis-Ooguri model are:

$$\begin{aligned} \partial_+ \partial_- X^{A'} &= 0, & \partial_- X^+ &= 0, & \partial_+ X^- &= 0, \\ \partial_- \lambda^- &= 0, & \partial_+ \lambda^+ &= 0. \end{aligned}$$

While the energy momentum tensor constraints become:

$$\begin{aligned} T_{++} &= -2\pi T \left(\partial_+ X^{A'} \partial_+ X_{A'} + \lambda^- \partial_+ X^+ \right) = 0, \\ T_{--} &= -2\pi T \left(\partial_- X^{A'} \partial_- X_{A'} + \lambda^+ \partial_- X^- \right) = 0. \end{aligned}$$

In order for the spectrum to be nontrivial, we compactify the longitudinal coordinate $X^1 \sim X^1 + 2\pi wR$. The dispersion relation then becomes [4]:

$$E = \frac{P^{A'} P_{A'}}{4\pi T w R} + \frac{N + \tilde{N} - 2}{w R}.$$

Along with the level matching condition $N - \tilde{N} = wn$. The conserved Noether charges of the Gomis-Ooguri string are:

$$\begin{aligned} P_{A'} &= \int d\sigma \Pi_{A'}, & P_A &= \int d\sigma \Pi_A, \\ J_{A'B'} &= 2 \int d\sigma \Pi_{[A'} X_{B']}, & M_{AB} &= 2 \int d\sigma \Pi_{[A} X_{B]}, \\ G_{A'}^A &= \int d\sigma (\Pi_{A'} X^A - T X_{A'} \epsilon^A{}_B \partial_\sigma X^B). \end{aligned}$$

Where Π_A and $\Pi_{A'}$ denote the canonical momenta. The non-vanishing commutation relations are then given by:

$$\begin{aligned} [P_{A'}, G_{B'}^A] &= \delta_{A'B'} Z^A, & [J_{B'C'}, G_{A'}^B] &= 2\delta_{A'[B'} G_{C']}^B, \\ [G_{A'}^A, G_{B'}^B] &= \delta_{A'B'} Z^{AB}, & [J_{A'B'}, J_{C'D'}] &= 4\delta_{[A'[C'} J_{D']}^{B]}, \\ [M_{AB}, Z_{CD}] &= 2\eta_{A[C} Z_{B]D}, & [G_{A'}^B, P_A] &= \delta_A^B P_{A'}, \\ [G_{A'}^A, P_{B'}] &= \delta_{A'B'} Z^A, & [M_{AB}, P_C] &= 2\eta_{[BC} P_{A]}, \\ [P_A, Z^{BC}] &= 2\delta_A^{[B} Z^{C]}, & [M_{AB}, G_{A'}^C] &= 2\delta_{[A}^C G_{|A'|B]}. \end{aligned}$$

This is called the string Newton-Cartan algebra [5, 6, 7] and the extensions Z^A and Z^{AB} are defined by:

$$\begin{aligned} Z^A &= T \int d\sigma \epsilon^A{}_B \partial_\sigma X^B, \\ Z^{AB} &= 2T \int d\sigma X^{[A} \epsilon^{B]}{}_C \partial_\sigma X^C. \end{aligned}$$

Lewis Carroll's 'Through the Looking-Glass'

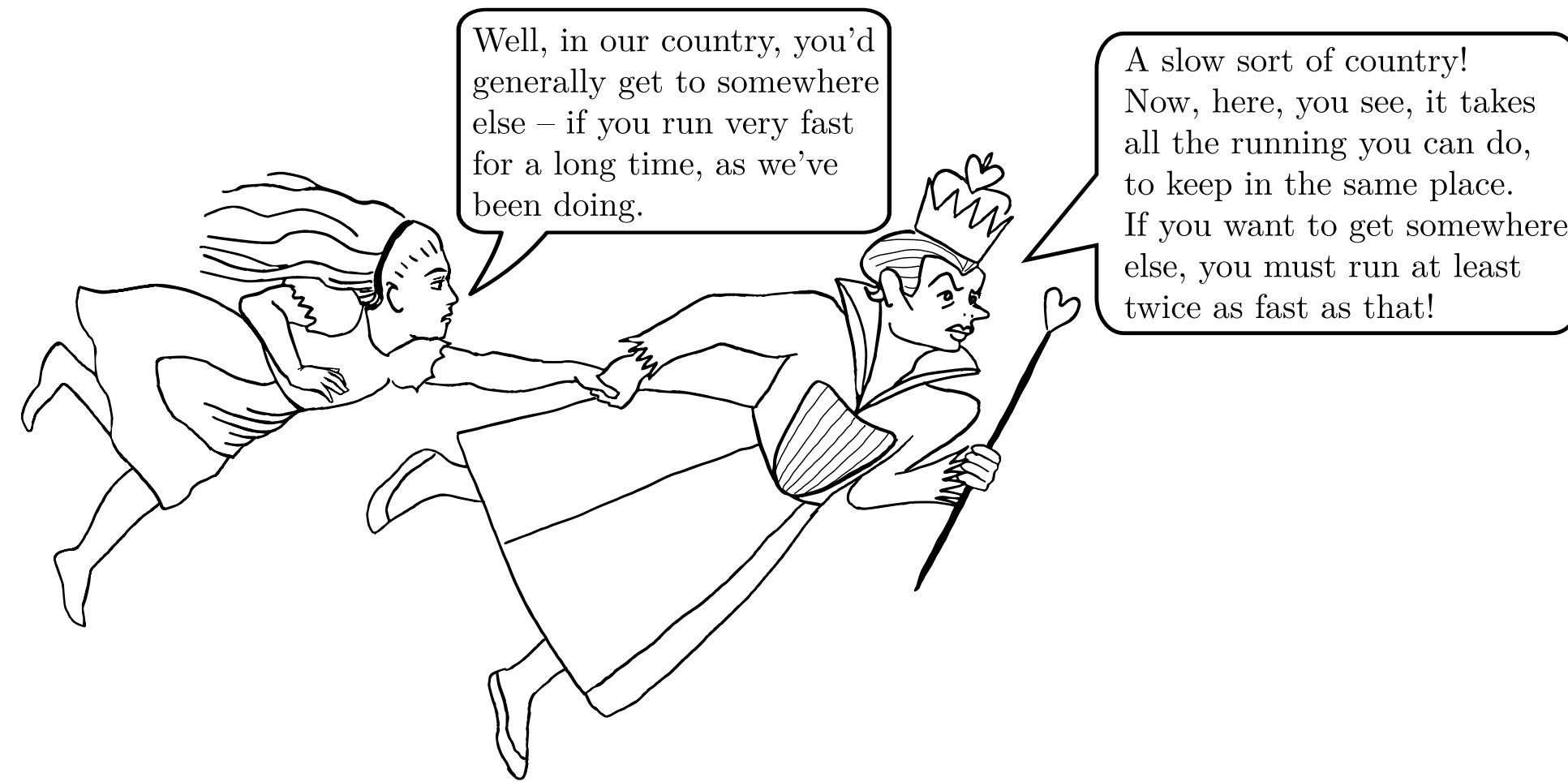


Illustration by   lfr  ur Edda Sigur  ard  ttir. Text in figure by Lewis Carroll [8].

Phase Space Carroll String (First Model)

In [9] a Carroll limit of the canonical phase-space Nambu-Goto action was considered. Starting from:

$$\hat{S}_{\text{NG}} = \int d^2\sigma \left(\hat{\Pi} \cdot \dot{\hat{X}} - \hat{\mu} \hat{\Pi} \cdot \hat{X}' - \frac{\hat{e}}{2} \left(\hat{\Pi}^2 + \hat{T}^2 \hat{X}'^2 \right) \right).$$

A way to take the *   la string* Carroll limit, as advertised in [9], is by the following rescaling:

$$\hat{X}^A = c X^A, \quad \hat{\Pi}^A = \frac{1}{c} \Pi^A, \quad \hat{\mu} = \mu, \quad \hat{e} = c^2 e, \quad \hat{T} = \frac{1}{c} T.$$

While $\hat{X}^{A'} = X^{A'}$ and $\hat{\Pi}^{A'} = \Pi^{A'}$. Taking $c \rightarrow 0$ this yields:

$$S_{\text{C}} = \int d^2\sigma \left(\Pi \cdot \dot{X} - \mu (\Pi \cdot X') - \frac{e}{2} \left(\Pi^A \Pi_A + T^2 X'^{A'} X'_{A'} \right) \right).$$

This action is invariant under Stringy Carroll symmetries, but it has trivial dynamics. In conformal gauge the equations of motion for the transverse coordinates are given by:

$$\partial_\tau X^{A'} = 0.$$

Hence, the transverse coordinates of this string are constant. The dispersion relation for this Carroll string is given by:

$$P^A P_A + T^2 \partial_\sigma X^{A'} \partial_\sigma X_{A'} = 0.$$

Chiral Carroll String (Second Model)

There is another way to take the aforementioned limit. Instead, let us start from the canonical phase-space Polyakov action:

$$S_P = \int d^2\sigma \left(\hat{\Pi} \cdot \dot{\hat{X}} + \frac{\gamma^{\tau\sigma} \hat{\Pi} \cdot \hat{X}'}{\gamma^{\tau\tau}} + \frac{1}{2T\sqrt{-\gamma}\gamma^{\tau\tau}} \left(\hat{\Pi}^2 + \hat{T}^2 \hat{X}'^2 \right) \right).$$

Let us now rescale in the following way:

$$\hat{X}^A = c X^A, \quad \hat{\Pi}^A = \frac{1}{c} \Pi^A, \quad \hat{T} = \frac{1}{c^2} T.$$

While $\hat{X}^{A'} = X^{A'}$, $\hat{\Pi}^{A'} = \Pi^{A'}$ and the components of the worldsheet metric, $\gamma^{\tau\sigma}, \gamma^{\tau\tau}, \gamma^{\sigma\sigma}$ do not scale with c . Therefore:

$$\begin{aligned} S_P &= \int d^2\sigma \left(\Pi \cdot \dot{X} + \frac{\gamma^{\tau\sigma} \Pi \cdot X'}{\gamma^{\tau\tau}} \right. \\ &\quad \left. + \frac{1}{2T\sqrt{-\gamma}\gamma^{\tau\tau}} \left(\Pi^A \Pi_A + c^2 \Pi^{A'} \Pi_{A'} + T^2 X'^A X'_A + \frac{T^2}{c^2} X'^{A'} X'_{A'} \right) \right). \end{aligned}$$

Integrating out the canonical momenta Π^A and $\Pi^{A'}$ gives:

$$S_P = -\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} \gamma^{ab} \left(\partial_a X^A \partial_b X_A + \frac{1}{c^2} \partial_a X^{A'} \partial_b X_{A'} \right).$$

At this stage we introduce auxiliary fields $\lambda_a^{A'}$ and $\bar{\lambda}_a^{A'}$ such that the divergent part is classically equivalent to

$$-\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} (\Gamma_+^{ab} \lambda_a^{A'} \partial_b X_{A'} + c^2 \Gamma_-^{ab} \bar{\lambda}_a^{A'} \partial_b X_{A'} - 2c^4 \gamma^{ab} \lambda_a^{A'} \bar{\lambda}_b^{B'} \delta_{A'B'}).$$

After taking the limit $c \rightarrow 0$ we find the Chiral Carroll string:

$$S_{\text{CC}} = -\frac{T}{2} \int d^2\sigma \sqrt{-\gamma} \left(\gamma^{ab} \partial_a X^A \partial_b X_A + \Gamma_+^{ab} \lambda_a^{A'} \partial_b X_{A'} \right).$$

The equations of motion in light-cone gauge are:

$$\partial_+ \partial_- X^A = 0, \quad \partial_- X^{A'} = 0, \quad \partial_- \lambda^{A'} = 0.$$

The transverse coordinates have nontrivial dynamics, but the dispersion relation of this action cannot be determined.

Extended Carroll String (Third Model)

Similarly we may consider the *Extended Carroll String* which we define by the following action:

$$S_{\text{C}} = T_{\text{eff}} \int d^2\sigma \left(\partial_+ X^A \partial_- X_A + \lambda_k^- \partial_- X_k^+ + \lambda_k^+ \partial_+ X_k^- \right).$$

Where we have defined transverse *Gomis-Ooguri pairs* by:

$$X_k^{\pm} = X^{2k} \pm X^{2k+1}, \quad \text{for } k \in \{1, \dots, N\}.$$

Where each even transverse coordinate X^{2k} is taken to be temporal, while the odd transverse coordinates X^{2k+1} are taken to be spatial. The equations of motion of the Extended Carroll String are:

$$\partial_+ \partial_- X^A = 0, \quad \partial_{\pm} X_k^{\mp} = 0, \quad \partial_{\pm} \lambda_k^{\pm} = 0.$$

While the energy momentum tensor constraints are given by:

$$T_{++} = -2\pi T_{\text{eff}} \left(\partial_+ X^A \partial_+ X_A + \lambda_k^- \partial_+ X_k^+ \right) = 0,$$

$$T_{--} = -2\pi T_{\text{eff}} \left(\partial_- X^A \partial_- X_A + \lambda_k^+ \partial_- X_k^- \right) = 0.$$

This action is invariant under the following Carroll symmetries:

$$\delta X^A = \xi^A + \Lambda^A{}_{A'} X^{A'} + \Lambda^A{}_B X^B,$$

$$\delta X^{A'} = \xi^{A'} + \Lambda^{A'}{}_{B'} X^{B'},$$

$$\text{along with } \Lambda^{2k+1}{}_{2l} - \Lambda^{2k}{}_{2l+1} = \Lambda^{2k}{}_{2l} - \Lambda^{2k+1}{}_{2l+1} = 0.$$

We find transverse central extensions, Z^{2k} and Z^{2k+1} arising in the following commutators:

$$[P_A, C_B^{2k}] = \eta_{AB} Z^{2k+1}, \quad [P_A, C_B^{2k+1}] = \eta_{AB} Z^{2k}.$$

Where P_A denotes the charge associated with longitudinal translations and C_B^{2k} and C_B^{2k+1} denote the charges associated with Carroll boosts from the longitudinal sector to the transverse temporal sector and transverse spatial sector respectively. Similarly we find extensions $Z^{2k}{}_{2l}$, $Z^{2k+1}{}_{2l+1}$ and $Z^{2k+1}{}_{2l}$ when computing the following commutators:

$$\begin{aligned} [C_A^{2k}, C_B^{2l}] &= \eta_{AB} Z^{2k}{}_{2l}, \\ [C_A^{2k+1}, C_B^{2l+1}] &= \eta_{AB} Z^{2k+1}{}_{2l+1}, \\ [C_A^{2k+1}, C_B^{2l}] &= \eta_{AB} Z^{2k+1}{}_{2l}. \end{aligned}$$

While the dispersion relation is given by:

$$P^A P_A + 2\pi T_{\text{eff}} \sum_{k=1}^N w_k R_k P_{2k+1} = 0.$$

Where we have assumed closed boundary conditions and we have compactified the even transverse coordinates:

$$X^{2k} \sim X^{2k} + 2\pi w_k R_k.$$

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